

Chashi Mahiul Islam

☎ +1 (850) 559-8366 • ✉ chashi.buet@gmail.com
🌐 <https://linkedin.com/in/chashi-bd> • 📄 <https://github.com/thechashi> • 🌐 <https://chashi.netlify.app/>

EDUCATION

Florida State University

Ph.D. Candidate in Computer Science (CS) (Advisor: Dr. Xiuwen Liu)

Tallahassee, FL

December, 2026 (Expected)

- Research Interest: Robustness of Large Multimodal Models

Florida State University

M.Sc. in Computer Science

Tallahassee, FL

August, 2025

- CGPA: 3.97/4.00

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

April, 2019

TECHNICAL SKILLS

Languages: Highly proficient-Python, C/C++, Javascript, SQL, HTML/CSS

Machine Learning Frameworks/Libraries: PyTorch, TensorFlow, Scikit-learn, HuggingFace, Transformers, CLIP, PEFT, Pandas, NumPy, NLTK, Matplotlib, Tableau, accelerate, Keras, OpenCV, XGBoost, LangChain, AutoGen, CrewAI, LlamaIndex, LangGraph

Tools: React, Angular, Flask, Spring Boot, AWS, Google Cloud, Docker, Git, VS Code

Databases/Systems: PostgreSQL, MongoDB, Redis, Linux, macOS

WORK EXPERIENCE

Software Engineer Intern - Meta, Seattle, WA

May 2025 – Aug 2025

- Optimized large-scale ranking systems by implementing feature selection techniques and developing scalable data pipelines to integrate new features, improving key metrics in feed ranking models.

Graduate Research Assistant - Dept. of CS, Florida State University, Tallahassee, FL

Aug 2024 – Ongoing

- Conducting research on robustness and explainability of multimodal AI systems, focusing on vision-language models, large language models, and applying machine learning techniques to enhance AI reliability and security.

Graduate Teaching Assistant - Dept. of CS, Florida State University, Tallahassee, FL

Jan 2021 – Jul 2024

- Led recitations, developed assessments, and provided student (60-90) support for various computer science courses, including Secure, Parallel & Distributed Programming, Data Structures, Advanced Python, and OOP with C++.

Software Developer - CoKreates Limited, Dhaka, Bangladesh

Aug 2019 – Dec 2020

- Front-end design and development for Admin Panel and Human Resource Module of ERP system. Implemented RESTful APIs for the backend of the HR module and central logging system. Stack: **Spring Boot, Angular**

SELECTED PUBLICATIONS

• **Islam, C.M., Jacob Chacko, S., Nishino, M., Liu, X. (2025). NeuroShield-ViT: Mechanistic Understandings of Representation Vulnerabilities and Engineering Robust Vision Transformers. (ISVC 2025, Oral).** To appear in Lecture Notes in Computer Science (LNCS), Springer.

• **Islam, C.M., Jacob Chacko, S., Horne, P., Liu, X. (2025). Vision Embeddings and Their Role in Hallucination Vulnerabilities of Multimodal Large Language Models. (ISVC 2025, Poster).** To appear in Lecture Notes in Computer Science (LNCS), Springer.

• **Islam, C.M., Salman, S., Shams, M., Liu, X., Kumar, P. (2024). Malicious Path Manipulations via Exploitation of Representation Vulnerabilities of Vision-Language Navigation Systems. (IROS 2024, Oral).** In IEEE/RSJ International Conference on Intelligent Robots and Systems.

RESEARCH EXPERIENCE

Neural Expansion: A Unified Mechanism for Deep Network Generalization (Manuscript ready to submit):

- Proposed Generalization Intervals framework explaining why more than 90% of correct predictions occur within directional stability regions defined by Jacobian singular values. ([codebase](#))

The Illusion of Precision: Numerical Instability of Large Language Models:

- Discovered microscopic input perturbations cause 10-million-fold amplification in LLMs, collapsing logit margins and triggering unpredictable token flips. ([codebase](#))

Robust Vision Transformers (ViTs) - Understanding and Mitigating Representation Vulnerabilities (ISVC 2025):

- Examined vulnerabilities in Vision Transformers (ViTs), revealing how small input changes affect layer-wise representations.
- Proposed NeuroShield-ViT to neutralize vulnerable neurons and enhance adversarial robustness.
- Improved classification accuracy by 54.85% against PGD attacks and 71.6% against IGO attacks on ImageNet1K.

Adversarial Exploitation in Robot Vision-Language Navigation (IROS 2024):

- Devised algorithms to adversarially modify images by exploiting semantic representation gaps in vision-language models.
- Redirected robot navigation with a 91.7% landmark matching rate, demonstrating effective adversarial manipulation.
- Enhanced security via detection mechanism for manipulated image noise, achieving 96.3% accuracy. ([codebase](#))

Enhancing Visual Spatial Reasoning through Multi-Task Learning (ISVC 2025):

- Developed SpatialViLT, a vision-language model with spatial features (depth, 3D, edge) via multi-task learning.
- Improved spatial understanding and boosted SOTA performance on the Visual Spatial Reasoning (VSR) dataset by 2.1%.

Leveraging Superresolution for Enhanced Analysis of Low-Resolution Content :

- Evaluated SOTA models for identifying low-resolution deep fakes in collected video data.
- Improved deep fake detection, boosting test accuracy by 19% on the FaceShifter dataset with Xception Net. ([codebase](#))

SELECTED PROJECTS

Reliability Limits of Deep De Novo Peptide Sequencing (Transformers, Pyteomics) - 2025:

- Found 73.6% and 95.8% prediction instability in Casanovo and π -PrimeNovo under physics-preserving spectral perturbations, revealing fundamental reliability gaps in de novo sequencing models. ([codebase](#))

Deciphering Poorly Formatted Text using Private LLM (Transformers, PEFT, BitsAndBytes) - 2023:

- Processed unstructured text using private LLMs hosted locally for data privacy, enhancing text processing efficiency.
- Applied PEFT (LoRA) for efficient finetuning of summarizers and classifiers, boosting performance with prompt engineering. ([codebase](#))

Multivariate Time Series Classification (Pytorch, Scikit-learn, tsfresh) - 2023:

- Built a predictive XGBoost model to forecast intracerebral hemorrhage using time-series data and PubMed radiology reports.
- Achieved 74.11% accuracy and 75.51% precision, aiding hospital decisions by identifying high-risk patients. ([codebase](#))

Comparison of Summarization Techniques Using Language Models (Transformers, NLTK) - 2023:

- Investigated transformer models for text summarization, comparing BERT, GPT-2, T5, and BART for medical reports.
- Addressed input size limits and domain-specific vocabulary, improving summarization effectiveness. ([codebase](#))

Web scraping: Higher Education Funding Offers (Selenium, Scikit-learn) - 2023:

- Addressed funding challenges for international students in US by developing a comprehensive web scraping pipeline. ([codebase](#))
- Used requests, Selenium, BeautifulSoup, and pandas to extract data on professors, from CSRank website, seeking new students.

Image Style Transfer with VGG Network (PyTorch) - 2021:

- Implemented image style transfer methodology using convolutional neural networks. ([codebase](#))
- Used a pretrained VGG 19 network to extract content and style representations, generating a new target image.

Sequence to Sequence Stock Time Series Data Prediction Model (PyTorch) - 2021:

- Developed an LSTM-based seq2seq model for stock prediction using Alpha Vantage API and PyTorch. ([codebase](#))
- Added CLI features for stock prediction, data plotting, and model saving, delivering accurate predictions with documentation.

RELEVANT COURSES

Graduate: Introduction to Data Science, Data Mining, Advanced Database, Advanced Algorithms, Analytic Methods, Deep Learning and Reinforcement Learning Fundamentals, Data and Computer Communication

Undergraduate: Python, C, C++, OOP, Digital Signal Processing, Computer Graphics, Software Development, Discrete Mathematics, Probability and Statistics, Database Management System, Image Processing and Pattern Recognition

LEADERSHIP AND AWARDS

Naaman Franklin Faile Jr. Graduate Fellowship: Scholarship for Fall 2025	2025
1st Runner-up: ACM at FSU Fall Programming Contest	2024
2nd Runner-up: ACM at FSU Fall Programming Contest	2023
Cultural Secretary: Bangladesh Student Association-FSU	2021
President: BUET Film Society	2018